```
# %% [markdown]
# # Week 1 — End-to-end toy dataset test
# Verifies: GeoNet FDSN access, ObsPy download, SeisBench + EQTransformer picking
# Toy day: 2019-09-04, the day of the M5.2 Lake Taupō mainshock — chosen delibera
# since it sits inside our revised 1 Aug-30 Sep 2019 analysis window, so this tes
# doubles as a first look at real data from the actual study period.
#
# Run this in VS Code connected to the HPC via Remote-SSH, with the `tvz-seismic`
# conda env selected as the Jupyter kernel (VS Code renders '# %%' blocks as cell

# %%
import obspy
from obspy.clients.fdsn import Client
from obspy import UTCDateTime
import os

OUTPUT_DIR = os.path.expanduser("~/tvz_project/data/toy_test")
os.makedirs(OUTPUT_DIR, exist_ok=True)

client = Client("GEONET")

t0 = UTCDateTime("2019-09-04T00:00:00")
t1 = UTCDateTime("2019-09-05T00:00:00")

# %%
# Pull station inventory for a small handful of stations near Lake Taupō first,
# to confirm which were live on this date before bulk-downloading waveforms.
inventory = client.get_stations(
    network="NZ",
    station="*",
    latitude=-38.8, longitude=176.0, maxradius=1.0,   # ~1 deg radius around Lake
    starttime=t0, endtime=t1,
    level="channel",
)
print(inventory)

# %%
# Pick 3 stations from the inventory result above and download waveforms.
# Replace these with real station codes once you've inspected `inventory`.
test_stations = [sta.code for net in inventory for sta in net][:3]
print("Testing with stations:", test_stations)

st = obspy.Stream()
for sta in test_stations:
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    try:
        st += client.get_waveforms(
            network="NZ", station=sta, location="*", channel="HH?",
            starttime=t0, endtime=t1,
        )
    except Exception as e:
        print(f"  {sta}: download failed — {e}")

print(st)
st.write(os.path.join(OUTPUT_DIR, "toy_2019-09-04.mseed"), format="MSEED")

# %%
# Full-stack test: run SeisBench's EQTransformer over the toy day.
import seisbench.models as sbm

model = sbm.EQTransformer.from_pretrained("original")
model.eval()

# Conservative pre-trained model, matching the proposal's parameters
picks = model.classify(
    st,
    batch_size=8,
    P_threshold=0.2,
    S_threshold=0.2,
)

print(f"Detected {len(picks.picks)} picks on toy day")
for p in picks.picks[:10]:
    print(p)

# %%
print("=== Week 1 toy test complete ===")
print("If picks were produced above, ObsPy -> SeisBench -> EQTransformer is verif
print("Next: NonLinLoc + GrowClust3D.jl link-up (Week 1 remaining task) and Week
print("station inventory download for the Aug-Sep 2019 window.")
```